EXPLORATORY DATA ANALYSIS

Task 5 • Data Analyst Internship

Dataset: Titanic passenger dataset (train.csv)
Purpose: Discover patterns, relationships, missing values, and key factors associated with passenger survival.

Dataset profile	Finding
Rows / columns	891 passengers / 12 variables
Overall survival rate	38.4% (342 survivors)
Missing Age values	177 (19.9%)
Missing Cabin values	687 (77.1%)
Missing Embarked values	2

Prepared as a concise, submission-ready EDA report with visual evidence and actionable observations.

1. Objective and Method

Exploratory Data Analysis (EDA) is the process of examining a dataset using descriptive statistics and visualizations before building models. This analysis uses Pandas for data inspection and aggregation, and Matplotlib for charts.

Workflow followed

1. Loaded the dataset and reviewed its dimensions and data types.
2. Checked missing values and duplicate records.
3. Summarized numerical variables with descriptive statistics.
4. Compared survival across sex and passenger class.
5. Reviewed age distribution and numeric correlations.
6. Recorded findings, limitations, and suggested next steps.

2. Data Quality Review

Column	Missing	Missing %
Age	177	19.9%
Cabin	687	77.1%
Embarked	2	0.2%

Age has 177 missing values and Cabin has 687; Embarked has 2 missing values. Cabin is highly incomplete (77.1%), so it should not be used naively without a missingness strategy. Duplicate rows: 0.

Recommended preparation: impute missing Age with a suitable group median or model-based estimate; fill the small number of missing Embarked values with the mode if justified; extract useful cabin-deck information before deciding whether to drop Cabin. Keep a record of every change.

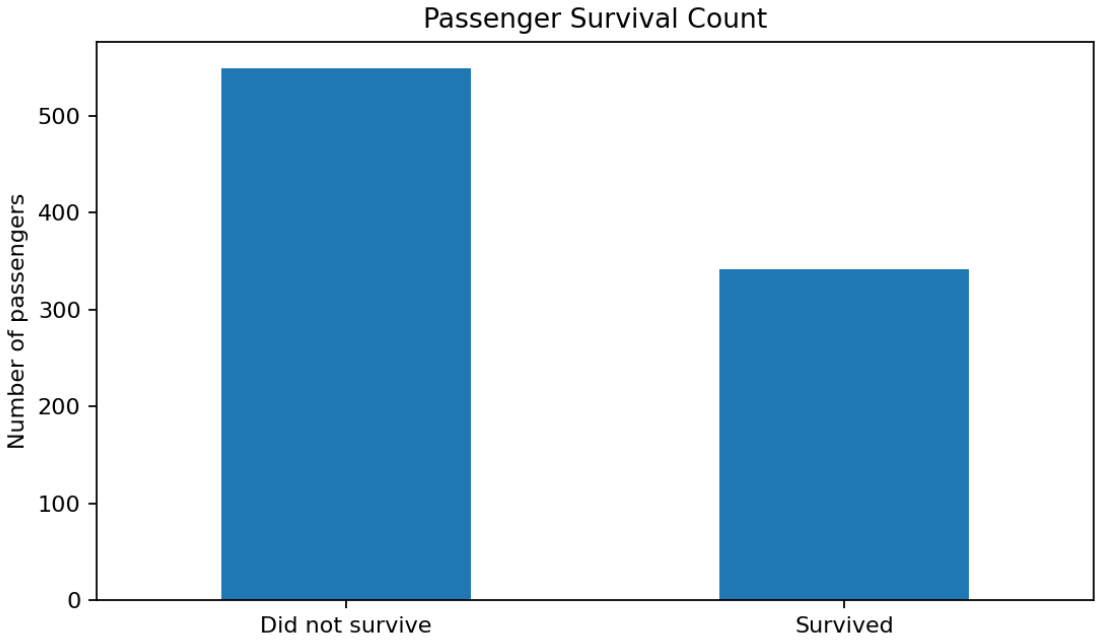
3. Descriptive Statistics

Statistic	Age	Fare	SibSp	Parch
count	714.00	891.00	891.00	891.00
mean	29.70	32.20	0.52	0.38
std	14.53	49.69	1.10	0.81
min	0.42	0.00	0.00	0.00
25%	20.12	7.91	0.00	0.00
50%	28.00	14.45	0.00	0.00
75%	38.00	31.00	1.00	0.00
max	80.00	512.33	8.00	6.00

The median passenger age is 28.0 years and the median fare is 14.45. Fare is typically right-skewed because a smaller number of passengers paid substantially higher fares.

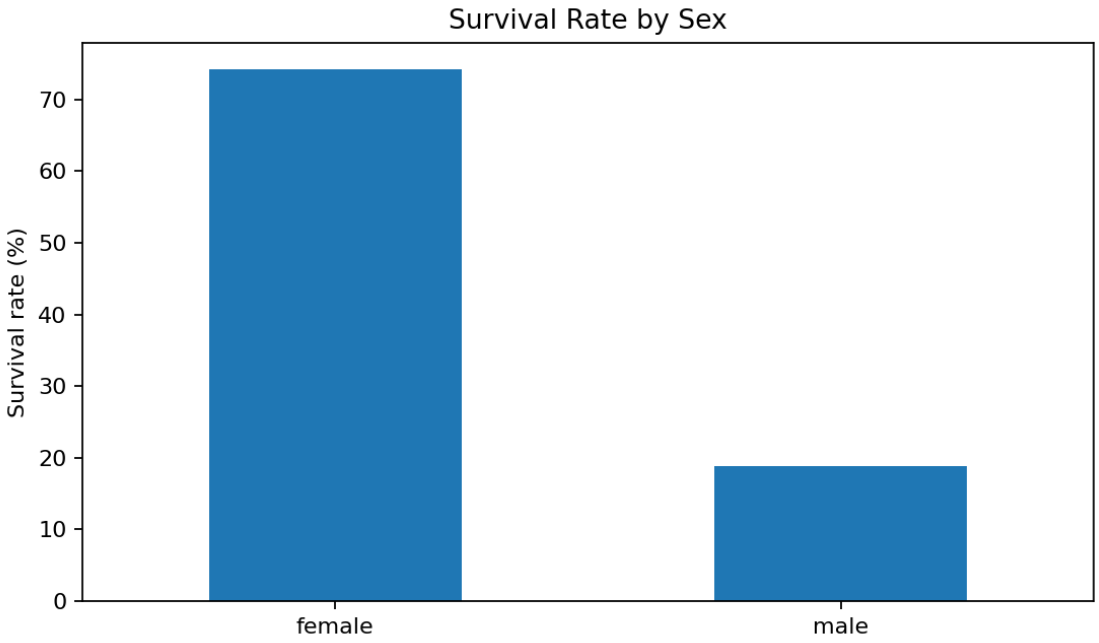
4. Visual Analysis and Observations

Chart 1 — Survival counts



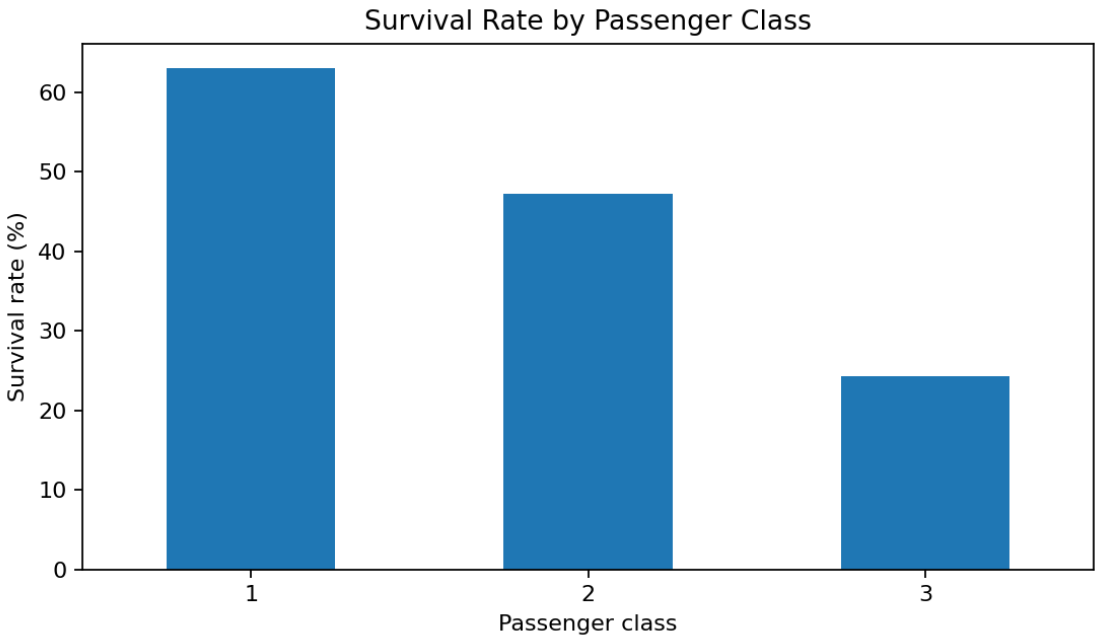
Observation: 549 passengers did not survive and 342 survived. The overall survival rate was 38.4%, so non-survivors were the larger group.

Chart 2 — Survival rate by sex



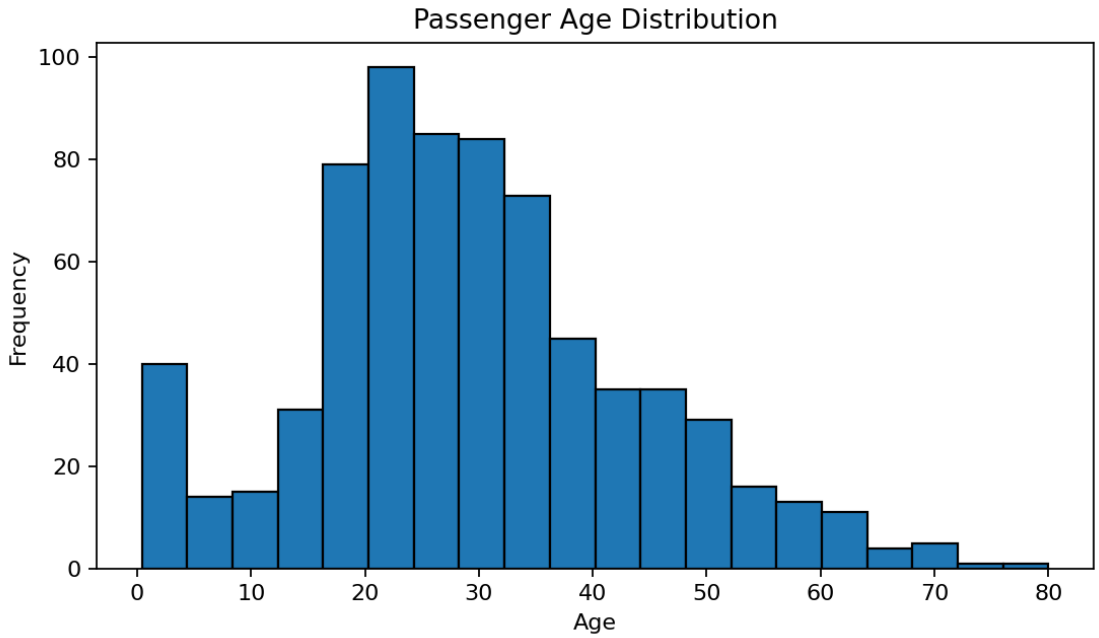
Observation: female survival was 74.2% compared with 18.9% for males. Sex shows a strong association with survival in this dataset; this is an observed relationship, not proof of causation.

Chart 3 — Survival rate by passenger class



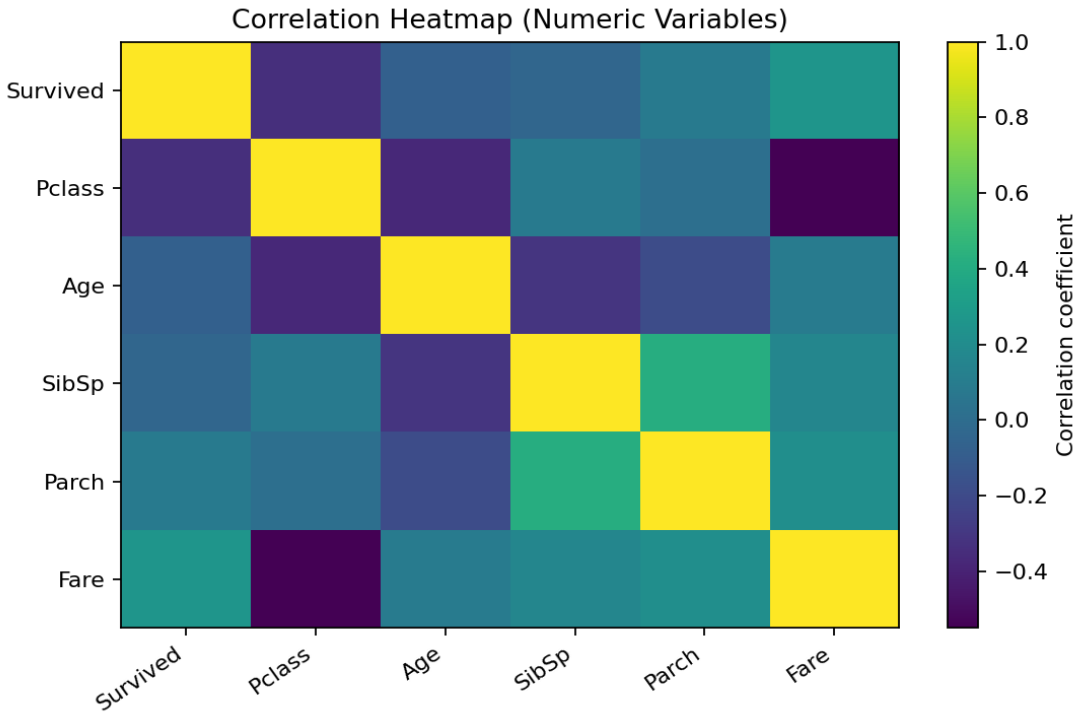
Observation: survival differs substantially across passenger classes. First-class passengers had the highest survival rate and third-class passengers the lowest, indicating passenger class is an important feature to examine further.

Chart 4 — Age distribution



Observation: most passengers were adults, with fewer passengers at older ages. Age contains missing values, so any age-based comparisons should state how missing ages were handled.

Chart 5 — Correlation heatmap



Observation: the heatmap summarizes linear relationships among selected numeric variables. Correlation does not establish causation, and categorical variables such as Sex and Embarked are not represented in this numeric-only heatmap.

5. Key Findings

1. The dataset contains 891 records and 12 columns.

2. Overall, 38.4% of passengers survived.

3. Female survival rate (74.2%) was much higher than male survival rate (18.9%).

4. Survival rate declined from first class (63.0%) to second class (47.3%) and third class (24.2%).

5. Age is missing for 177 passengers, while Cabin is missing for 687; these require explicit handling.

6. Fare and family-related variables (SibSp and Parch) may provide additional explanatory value and should be explored with grouped comparisons.

6. Conclusion and Next Steps

EDA reveals clear differences in survival by sex and passenger class, while also identifying data-quality issues that must be handled before further analysis. The next step is to document a repeatable cleaning process, create features such as family size and cabin deck where appropriate, and test a classification model using a train/test split. Any model should be evaluated with suitable metrics and checked for overfitting.

Tools used: Python, Pandas, Matplotlib. Dataset source supplied for this task: train.csv (Titanic dataset).